Reg. No. | | | | | | | | | | | [13ME652/12ME652]

SIXTH SEMESTER B.Tech./B.E. EXAMINATION, JUNE – 2016

QUALITY ASSURANCE AND RELIABILITY ENGINEERING

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data, if any, may be suitably assumed.
3. Use of statistical tables permitted.

1. a. Define process capability and list various charts available for variables with control limits. **08**

b. Thirty samples of size 6 yielded $\bar{\bar{x}} = 480$ and $\bar{R} = 34$. Compute control limits for $\bar{x}$ and R charts and estimate the standard deviation of the process. **06**

2. a. Briefly classify attribute control charts and write the procedure to be followed to construct P-chart. **08**

b. In a manufacturing unit of certain generators, the units are required to meet number of specifications, with which they are produced and subjected to inspection daily. At the end, 230 units have been rejected out of 4150 produced and inspected. **06**

i. Determine the 3σ control limits on a daily process.

ii. Only one point has crossed the control limits on that day 30 non confirming items were found in 200 units inspected. What value of P′ and control limits would you recommend for the future.

3. a. Define sampling plan and explain the differences between single, double and multiple sampling plans. **08**

b. In a double sampling plan N=5000, n_1=100, C_1=0, n_1+n_2=200, C_2=2. Compute Pa for 1% defective lot. **06**

4. a. Define and briefly explain producer's risk and consumer's risk. **04**

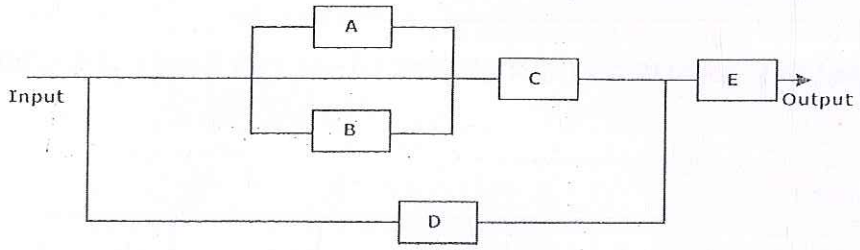
b. Write a short notes on **10**

i. AQL

ii. LTPO

5. a. Define reliability and explain briefly with the help of neat sketch 'failure rate curve'. **08**

b. Calculate the reliability of the configuration given below. **06**



Where $R_A = 0.7$, $R_B = 0.7$, $R_C=0.9$, $R_D=0.8$, and $R_E=0.9$

6. a. A cassette player has got 4 subsystems namely speed control system, the magnetic picking head, amplification and sound system with other systems. For satisfactory performance of a cassette player, all the 4 subsystems must perform satisfactorily. The MTBF of the various subsystems are: **08**

$MTBF_{scs}$ = 2000 hrs., $MTBF_{MPSS}$ = 2500 hrs, $MTBF_{A \text{ and } s}$ = 4000 hrs, and $MTBF_{OS}$ = 3000 hrs.

Calculate the reliability of the cassette player for 1500 hrs reception time. What is the mean time between failures of the cassette player?

b. Briefly explain with suitable examples – Reliability Management and Reliability Data Acquisition. **06**

7. a. Briefly explain with sketch "Fault tree analysis" with respect to reliability. **06**

b. What is hazard model and explain briefly. **08**

i. Constant hazard model

ii. Linear hazard model

8. a. Define unit redundancy **02**

b. Write short notes on any three: **12**

i. Weakest link technique

ii. Redundancy of optimization

iii. Reliability utility cost

iv. Mixed redundancy